MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

31 July 2026

Last completed milestone: M5 – Static equilibrium existence and uniqueness
Status: COMPLETE – AMBER
Next milestone: M6 – Unemployment flow balance – **NOT STARTED**

Source of truth. `docs/milestone_tracker.md`. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 through M4 are complete and green. M5 is complete and amber: uniqueness is green and existence is conditional on an additional bracket. M5b is an unstarted foundational refinement; M6 remains the next paper milestone.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium
Status: COMPLETE – GREEN
Paper: Sections 2–3; equations (1)–(7)
Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.
Expected declarations: `Primitives`; four assumption bundles; `cdf`; meeting rates; `surplus`; `continuationTerm`; `ValueEquilibrium`
Adequacy grade: GREEN **Dependencies:** None
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation
Status: COMPLETE – GREEN
Paper: Section 3, journal p. 401; equation (8)
Economic target: Derive equation (8) from equations (3)–(7).
Expected declarations: Continuation decomposition; `firm_share`; `surplus_flow_equation`; `surplus_bellman_of_probability`; `surplus_bellman`
Adequacy grade: GREEN **Dependencies:** M0 complete
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed conditional representation theorem.

M2. Affine surplus and endogenous reservation cutoff
Status: COMPLETE – GREEN
Paper: Section 3; reservation discussion and equation (12)
Economic target: Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).
Expected declarations: Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem
Adequacy grade: GREEN **Dependencies:** M1
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve M2 interfaces and design the measure-form continuation identity for M3.

M3. Continuation value, job destruction, and job creation
Status: COMPLETE – GREEN
Paper: Section 3; equations (9), (10), and (13)
Economic target: Measure-form continuation identity and derivation of job destruction and job creation.
Expected declarations: `expectedExcess`; `tailOptionValue`; equations (9), (10), and (13); JD/JC residual predicates; M3 capstone
Adequacy grade: GREEN **Dependencies:** M2
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-30 human mathematical/economic review; preserve the independent JD/JC interfaces and neutral capstone module.

M4. Reduced-equilibrium reconstruction and equivalence
Status: COMPLETE – GREEN
Paper: Section 3; equations (10), (13); journal p. 402
Economic target: Two-way representation between primitive values and the admissible reduced pair.
Expected declarations: `ReducedEquilibrium`; generic tail identity; forward and reverse maps; round trips; nonempty equivalence
Adequacy grade: GREEN **Dependencies:** M3
Blocker: None
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-31 human mathematical/economic review; preserve representation interfaces and formulate corrected continuity, endpoint, and crossing assumptions for M5.

M5. Existence and uniqueness of static equilibrium
Status: COMPLETE – AMBER
Paper: Section 3; equations (10), (13), Figure 1, journal p. 402
Economic target: Existence and uniqueness of the static reduced equilibrium.
Expected declarations: Expected-excess regularity; JD/JC slopes; `StaticExistenceAssumptions`; conditional unique existence; value-side economic uniqueness
Adequacy grade: Uniqueness: GREEN; existence: AMBER; overall: AMBER **Dependencies:** M4
Blocker: Existence uses a lower positive crossing condition not derived from primitives
Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-31 human review; preserve the conditional theorem and see the M5b design in `docs/static_existence_foundation.md`. This does not block M6.

M5b. Primitive foundation for static existence
Status: NOT STARTED
Paper: Foundational refinement to Section 3 static existence
Economic target: Replace the assumed lower crossing with a derived result.
Expected declarations: Proposed matching-boundary bundle; upper-job profitability; derived `lower_crossing`; optional Cobb–Douglas certification
Adequacy grade: Target: existence GREEN under a strengthened general theorem or certified Cobb–Douglas specialization **Dependencies:** M5
Blocker: The Inada condition on q alone is insufficient; JD-implied tightness must become positive through a separate profitability condition
Branch/commit or tag; last review; next action: No branch/tag or review; add right-hand Inada and upper-job profitability, derive `StaticExistenceAssumptions`, and see `docs/static_existence_foundation.md`. M5b is not M6.

M6. Unemployment flow balance
Status: NOT STARTED
Paper: Section 3; equation (14)
Economic target: Derive steady-state unemployment and define the full steady state.
Expected declarations: Flow balance; unemployment identity; full steady-state structure
Adequacy grade: – **Dependencies:** M4–M5
Blocker: Static equilibrium foundation
Branch/commit or tag; last review; next action: No branch/tag or review; introduce unemployment stocks only here.

M7. Global comparative statics
Status: NOT STARTED
Paper: Section 3
Economic target: Order-theoretic comparative statics for p, b, λ, r, σ where possible without differentiability.
Expected declarations: Parameter-order theorems; equilibrium order lemmas
Adequacy grade: – **Dependencies:** M5
Blocker: Global solution selection and parameter monotonicity
Branch/commit or tag; last review; next action: No branch/tag or review; separate global monotone results from derivatives.

M8. Appendix derivative comparative statics
Status: NOT STARTED
Paper: Appendix; equations (A1)–(A12)
Economic target: Differentiable equilibrium paths and derivative sign results.
Expected declarations: Analytic assumption layer; Jacobian/implicit-path theorems
Adequacy grade: – **Dependencies:** M5, M7
Blocker: Differentiability and nondegenerate Jacobian
Branch/commit or tag; last review; next action: No branch/tag or review; keep Appendix assumptions out of the core.

M9. Two-state anticipated cyclical shocks
Status: NOT STARTED
Paper: Section 4; equations (15)–(31)
Economic target: Two-state surplus systems, cutoff ordering under explicit hypotheses, and impact asymmetry.
Expected declarations: Two-state value system; ordering theorems; employment-measure impact results
Adequacy grade: – **Dependencies:** M3–M6
Blocker: Piecewise continuation and employment measure
Branch/commit or tag; last review; next action: No branch/tag or review; begin with an explicit two-state process.

M10. Finite Markov equilibrium and employment accounting
Status: NOT STARTED
Paper: Analytical prelude to Section 5; equations (32)–(38)
Economic target: Finite-state equilibrium representation, employment transition operator, creation/destruction accounting, and equation (38).
Expected declarations: Finite kernel; transition operator; mass preservation; accounting identity
Adequacy grade: – **Dependencies:** M6, M9
Blocker: Finite-state interface and numerical-solver connection
Branch/commit or tag; last review; next action: No branch/tag or review; prefer finite state before general measurable states.

NUM. Numerical simulation and calibration
Status: DEFERRED / OUTSIDE CORE LEAN
Paper: Section 5; equations (39)–(42), Tables I–II
Economic target: External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.
Expected declarations: External solver and optional Lean residual/accounting certificates
Adequacy grade: – **Dependencies:** Analytical interfaces as needed
Blocker: Numerical/empirical work is outside core Lean
Branch/commit or tag; last review; next action: No branch/tag or review; define a reproducible calibration and residual protocol externally.

M0 acceptance checklist

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U, not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

M1 acceptance checklist

☒ Generic continuation decomposition is proved under a probability measure.
☒ Firm surplus share is derived from equations (3) and (4).
☒ The surplus flow equation is derived from equations (5)–(7).
☒ Equation (8) is proved for the actual equilibrium surplus.
☒ The main theorem uses only probability normalization and existing integrability.
☒ The `ShockAssumptions` wrapper installs the probability instance locally.
☒ No M0 structure was strengthened or modified.
☒ No cutoff, affine surplus, or reduced equilibrium was introduced.
☒ Targeted builds pass.
☒ Full build passes.
☒ No `sorry`, `admit`, or custom axioms exist.
☒ Proof ledger TeX/PDF compiles and passes visual inspection.
☒ Milestone tracker TeX/PDF compiles and passes visual inspection.
☒ Human mathematical/economic review is complete.

M2 acceptance checklist

☒ Equation (8) is subtracted at two states.
☒ The scaled two-point surplus identity is proved.
☒ The divided affine-surplus identity is proved.
☒ Surplus is proved strictly increasing.
☒ The cutoff is defined from actual equilibrium surplus.
☒ The cutoff is proved to be a surplus zero.
☒ The zero is proved unique.
☒ $V = 0$ is derived from free entry and $r > 0$.
☒ $q(\theta)J(\varepsilon_u) = c$ follows from (1)–(2).
☒ $J(\varepsilon_u) > 0$ is proved.
☒ $S(\varepsilon_u) > 0$ is proved.
☒ $\varepsilon_d < \varepsilon_u$ is proved.
☒ Equation (12) is proved.
☒ Surplus signs are characterized relative to ε_d.
☒ Firm-value signs are characterized relative to ε_d.
☒ Active surplus is rewritten as a scaled positive part.
☒ No M0 or M1 mathematical declaration was changed.
☒ No M3 or later theorem was introduced.
☒ Targeted builds pass.
☒ Full build passes.
☒ No `sorry`, `admit`, or custom axioms exist.
☒ Proof-ledger PDF compiles and passes visual inspection.
☒ Tracker PDF compiles and passes visual inspection.
☒ Human mathematical/economic review is complete.

M3 acceptance checklist

- ☒ Expected excess is the integral of the positive part.
- ☒ Tail option value is defined from the induced CDF.
- ☒ Expected-excess integrability is proved.
- ☒ Strict upper-tail probability equals one minus the CDF.
- ☒ The strict-tail layer-cake identity is proved.
- ☒ Active-surplus expectation is rewritten using expected excess.
- ☒ Measure-form equation (9) is derived.
- ☒ Paper-facing equation (9) is derived.
- ☒ The free-entry search-gain identity is derived.
- ☒ Measure-form equation (10) is derived.
- ☒ Paper-facing equation (10) is derived.
- ☒ Job-destruction residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-destruction predicate.
- ☒ The product-form job-creation identity is derived.
- ☒ Paper equation (13) is derived.
- ☒ Job-creation residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-creation predicate.
- ☒ No `ReducedEquilibrium` structure is introduced.
- ☒ No M0–M2 mathematical declaration is changed.
- ☒ No M4 or later theorem is introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M4 acceptance checklist

- ☒ Robust product-form job-creation predicate is defined.
- ☒ Reduced pair stores positive theta and an admissible cutoff.
- ☒ Measure-form JD and product-form JC are the only stored equations.
- ☒ Generic positive-part integrability and tail identity are proved.
- ☒ Reduced paper equations (10) and (13) are derived.
- ☒ Every value equilibrium maps to a reduced equilibrium.
- ☒ Candidate surplus, J, U, W, V, and explicit Nash wage are defined.
- ☒ Active surplus is integrable and candidate equation (8) is proved.
- ☒ Reconstructed equations (1)–(7) are proved.
- ☒ A reduced pair reconstructs a value equilibrium and its cutoff.
- ☒ Reconstruct–reduce is exact; reverse economic components agree.
- ☒ Conditional nonempty equivalence is proved.
- ☒ No unconditional existence, uniqueness, M0–M3 change, or M5 theorem.
- ☒ Targeted and full builds pass; audits and PDFs pass.
- ☒ Human mathematical/economic review is complete.

M5 acceptance checklist

- ☒ Expected excess is antitone, one-Lipschitz, and continuous.
- ☒ JD net value and JD-implied tightness are defined.
- ☒ JD satisfaction is equivalent to $\theta = \theta_{JD}(d)$.
- ☒ The JD curve is strictly increasing.
- ☒ The scalar JC residual is defined and strictly decreasing on its admissible domain.
- ☒ Figure 1 JD-upward and JC-downward slope theorems are proved.
- ☒ At most one `ReducedEquilibrium` is proved.
- ☒ `StaticExistenceAssumptions` contains no root or equilibrium witness.
- ☒ Bracket continuity and upper residual $-c$ are proved.
- ☒ An interior root and a reduced witness are constructed by the IVT.
- ☒ Conditional unique reduced existence is proved.
- ☒ Value nonemptiness, theta/cutoff uniqueness, and economic uniqueness follow through M4.
- ☒ Existence is labeled conditional on the additional bracket assumptions.
- ☒ No M0–M4 theorem statement changed and no M6 theorem was introduced.
- ☒ Targeted/full builds, audits, and both PDFs pass.
- ☒ Human mathematical/economic review is complete.

M5b foundational follow-up

Status: NOT STARTED. The adequacy target is to upgrade static existence from amber to green under either a strengthened general matching-boundary theorem or a fully certified Cobb–Douglas specialization. M5b is a foundational refinement, not M6, and is not part of the current M5 Lean proof.

The proposed work adds a right-hand Inada condition for q, separately requires $b < p + \sigma\varepsilon_u$, proves that the JD curve reaches zero and then positive tightness, derives `lower_crossing`, and constructs `StaticExistenceAssumptions`. The Inada condition on q alone is insufficient because it says nothing about whether JD-implied tightness ever becomes positive. The complete design is in `docs/static_existence_foundation.md`. M6 remains **NOT STARTED**.

Changelog

2026-07-31 – M5 static-existence foundation documented. Recorded the exact current continuity and lower-crossing assumptions and the unimplemented M5b route combining right-hand Inada behavior with separate upper-job profitability. The note makes explicit that Inada alone is insufficient. M5 remains **COMPLETE – AMBER**; M6 remains **NOT STARTED**. The durable design note is `docs/static_existence_foundation.md`.

2026-07-31 – M5 review completed. The uniqueness and IVT proofs passed review. Figure 1 establishes uniqueness, not existence. The non-circular `lower_crossing` field assumes a positive residual rather than a root, but supplies the missing endpoint condition. Existence-only signatures now omit matching assumptions. Uniqueness is **COMPLETE – GREEN**; existence and overall M5 are **COMPLETE – AMBER**. A future task is to derive the lower crossing from primitive conditions on q, or certify it for Cobb–Douglas; this does not block M6.

2026-07-31 – M5 static existence and uniqueness implemented. Expected excess is one-Lipschitz; JD slopes upward and JC downward, proving at most one reduced equilibrium. The separate continuity/lower-bracket bundle and the derived upper residual $-c < 0$ yield an interior crossing by the IVT. M4 transports conditional existence and economic uniqueness to value equilibria. The implementation was submitted for review; equation (14) remains M6.

2026-07-31 – M4 review completed. Review found the generic excess and CDF-tail analysis valid, and equations (10) and (13) sufficient for reconstruction under the stated admissibility assumptions. The Nash wage is economically correct; equations (1)–(7) are reconstructed; and both economic round trips are certified. The import diagram and assumption-field account were corrected. M4 proves no existence or uniqueness theorem and is **COMPLETE – GREEN**.

2026-07-30 – M4 reconstruction and equivalence implemented. Created the robust reduced pair, generic analytic layer, forward bridge, explicit reverse reconstruction, and equivalence module. Equations (1)–(7) are reconstructed with the Nash wage; the reduced round trip is exact and the value round trip is componentwise exact. Nonemptiness equivalence consumes a witness on either side and proves no existence or uniqueness. M4 is **READY FOR REVIEW**.

2026-07-30 – M3 review completed. The layer-cake/CDF proof was reviewed and found valid without atomlessness. Equations (9), (10), and (13) have the correct signs and dependencies, and job creation is independent of the job-destruction equation. The capstone was moved to the neutral combining module `SteadyState/StaticConditions.lean`. Residual predicates require separate admissibility fields in M4, which must also prove a generic expected-excess/tail theorem for reverse reconstruction. M3 was graded **COMPLETE – GREEN**.

2026-07-30 – M3 continuation and static conditions implemented. Created `SteadyState/Continuation.lean`, `SteadyState/JobDestruction.lean`, and `SteadyState/JobCreation.lean`, with the capstone in the neutral `SteadyState/StaticConditions.lean`. Expected excess follows from the M2 active-surplus formula; Mathlib's strict-tail layer-cake theorem gives the CDF representation without atomlessness. Equations (9), (10), and (13), the two residual predicates, and the forward M3 capstone are proved for an existing `ValueEquilibrium`. No reduced-equilibrium structure or M4 theorem was introduced.

2026-07-29 – M2 review completed. Review confirmed that equation (8) was correctly subtracted at two states and that actual equilibrium surplus is affine and strictly increasing. The derived `reservationCutoff` is its uniquely identified zero; equations (1)–(2) establish $\varepsilon_d < \varepsilon_u$; and equation (12) and the reservation rule are proved. M2 remains conditional on an existing `ValueEquilibrium` and was graded **COMPLETE – GREEN**.

2026-07-29 – M2 affine surplus and cutoff implemented. Created `SteadyState/AffineSurplus.lean` and `SteadyState/Cutoff.lean`. Subtracting equation (8) gives affine surplus and strict monotonicity. The derived cutoff is the unique surplus zero; equations (1)–(2), positive vacancy cost and meeting, and $\beta < 1$ separately place it below ε_u. Added equation (12), reservation sign rules, the active-surplus bridge, and the M2 capstone. M2 is **READY FOR REVIEW**.

Review-archive convention. Beginning with M3, project-local review archives are named `review_m3.zip`, `review_m4.zip`, and so on. They remain untracked and are never committed.

2026-07-29 – M1 review completed. Human mathematical and economic review found equation (8) mathematically faithful to the paper and its proof non-circular. It is derived from equations (3)–(7), uses the actual equilibrium surplus, and assumes no cutoff, affinity result, or reduced equilibrium. It is a conditional representation theorem for an existing `ValueEquilibrium`; M1 does not prove equilibrium existence. M1 was graded **COMPLETE – GREEN**.

2026-07-29 – M1 equation (8) implemented. Created `MP1994V2/SteadyState/Surplus.lean` with the continuation decomposition, firm share, surplus flow equation, minimal probability theorem, and `ShockAssumptions` wrapper. Updated `All`, `Audit`, `README`, trackers, and added the proof ledger. M1 is **READY FOR REVIEW**.

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules → `All` → `Audit`; required explicit `ShockAssumptions P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build` certify `MP1994V2.All`.